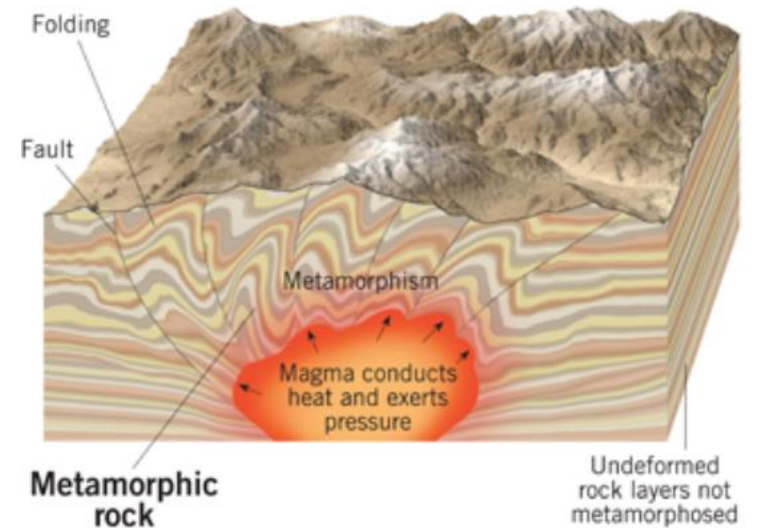
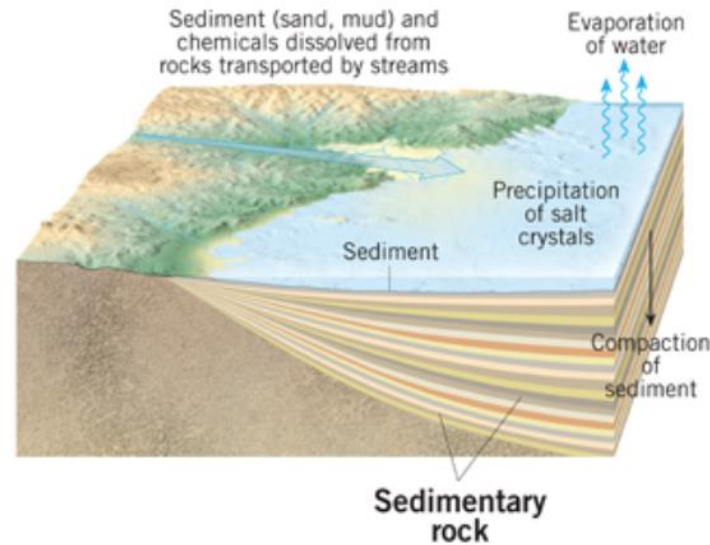
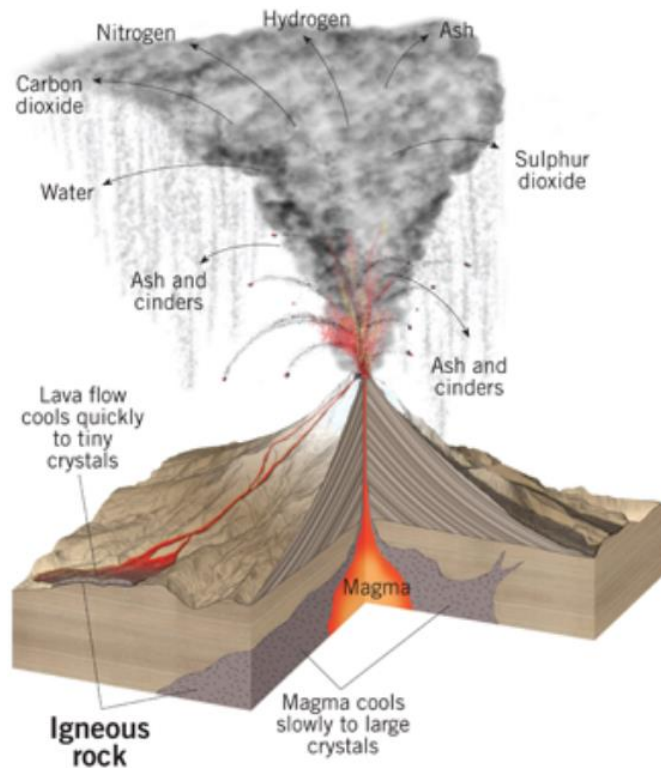


Virtual Rock Kit

Overview

- Geologists divide rocks into three major groups: igneous, sedimentary, and metamorphic



Overview

- Despite the difference in how these three rock types form, they are identified the same way – by characterizing their:
 - Texture – the overall appearance of a rock based on the size, shape, and arrangement of its component(s)
 - Composition – the components (minerals and/or clasts) that make up the rock

Objectives

- Characterize the texture and composition of igneous, sedimentary, and metamorphic rocks in your virtual rock kit;
- Identify their proper rock name using the provided identification tables.
 - The rock names highlighted and in bold uppercase font on the provided identification tables are provided in this virtual rock kit:

EXAMPLE

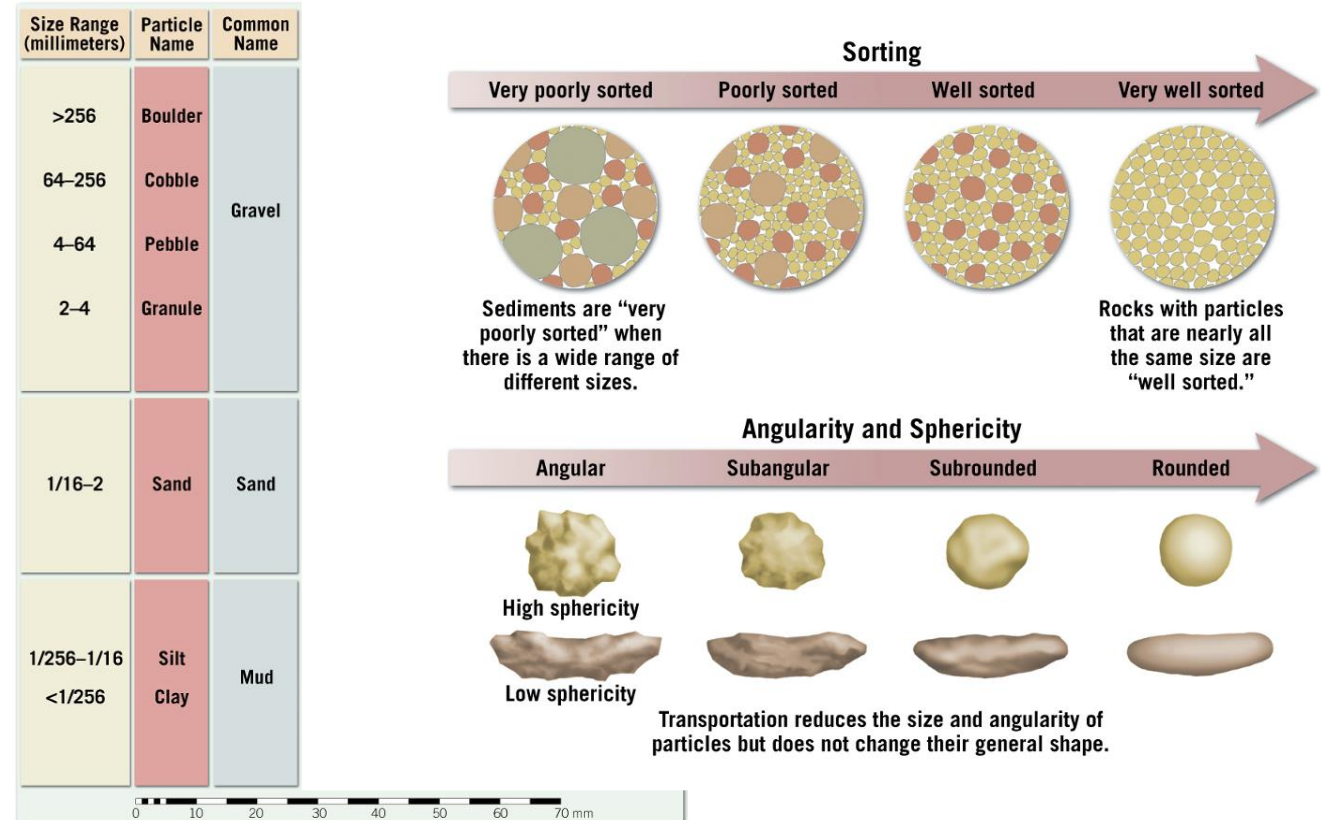
Sedimentary Rocks

Sedimentary Rock Identification

- Three categories of sedimentary rocks:
 - Clastic
 - Chemical
 - Biogenic

Sedimentary Rock Identification

- Clastic
 - Form from the accumulation of sediment and rock fragments transported away from original rock source
 - Categorized by texture
 - Size of sediment clasts
 - Shape of clasts
 - Relative sorting of clast sizes



Sedimentary Rock Identification

- Chemical
 - Form from saturated solution of minerals (mineral crystals grow in place)
 - Identified based on mineral composition
 - Common chemical sedimentary rocks composed of:
 - Quartz
 - Calcite
 - Gypsum
 - Halite

Sedimentary Rock Identification

- Biogenic
 - Identified by organic components
 - Form in place and made of organisms (plants/animals)
 - Organisms (shells, corals, microfossils)
 - Plant fragments (twigs, leaves, whole plants)

Sedimentary Rock Identification

Rock Texture and other distinctive properties		Rock Name
Clastic	Mostly angular gravel size grains (>2mm)	Breccia
	Mostly rounded gravel size grains (>2mm)	CONGLOMERATE
	Mostly sand size grains (1/16 to 2mm)	SANDSTONE
	No visible grains; mostly clay; layering visible	SHALE

Rock Texture and other distinctive properties		Rock Name
Biogenic	Black, layered, brittle rock; fossil plant fragments may be visible	Coal
	Mostly gravel-sized shells and shell or coral fragments	COQUINA
	Silty, earthy rock comprised of the microscopic shells of microfossils	Chalk

Rock Texture and other distinctive properties		Rock Name
Chemical	Composed of calcite; may have cavities, pores, or color banding; light in color	Travertine
	Composed of gypsum; gray, white, or colorless; can be scratched with fingernail	ROCK GYPSUM
	Composed of quartz; microcrystalline; hard (scratches glass)	CHERT

Sedimentary Rock Sample “G”



Clastic	✓
Chemical	
Biogenic	

Sedimentary Rock Sample “H”



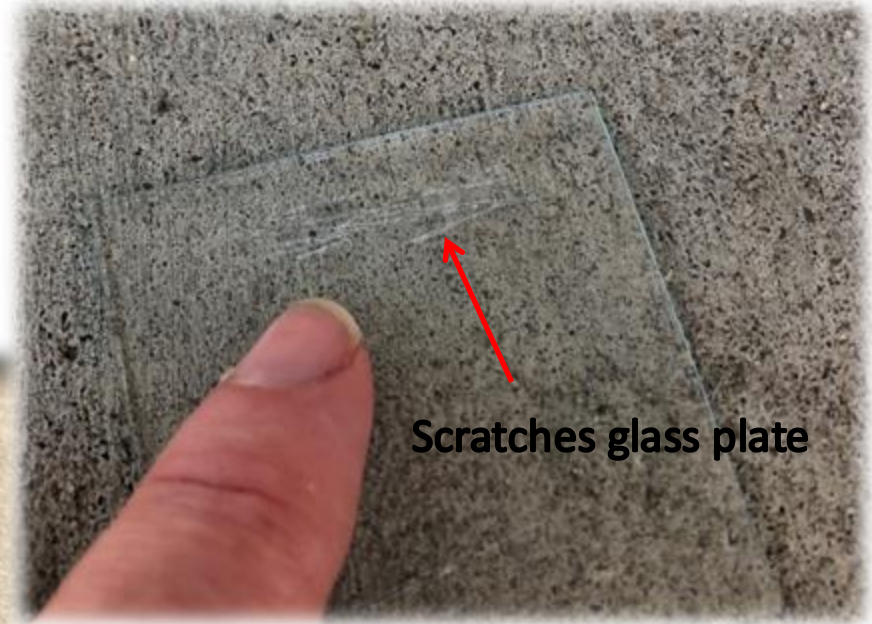
Clastic	✓
Chemical	
Biogenic	

Sedimentary Rock Sample “I”



Clastic	
Chemical	
Biogenic	✓

Sedimentary Rock Sample “J”



Clastic	
Chemical	✓
Biogenic	

Sedimentary Rock Sample “K”



Clastic	✓
Chemical	
Biogenic	

Sedimentary Rock Sample “L”



Clastic	
Chemical	✓
Biogenic	